

Track-gen yield study

Pure-Warp track-generation pipeline -- E=8192 yield study

device: cuda regime: half_width=0.5 m, scale=10 (1 m track / ~20 m box)

pipeline: generate -> resample -> XPBD relax -> inflate (all NVIDIA Warp)

Sweep results (end-to-end generate_tracks_warp, warmed + synced, E=8192):

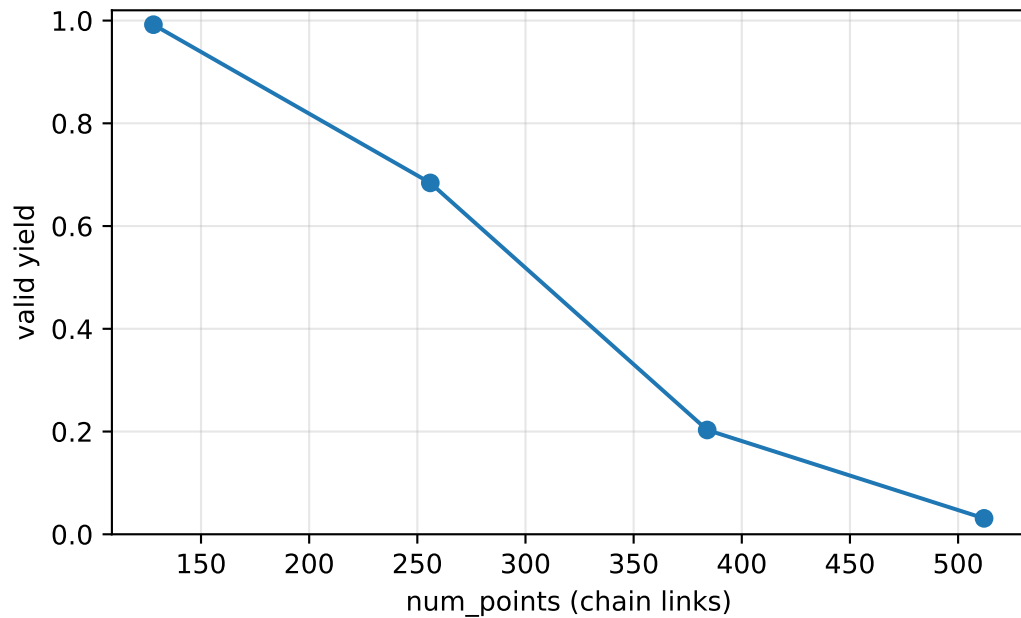
lever	yield	s/8192
-----	-----	-----
baseline 256 links/150 it	0.684	0.782
chain links (iters=150):		
128 links	0.992	0.213
256 links	0.684	0.782
384 links	0.203	1.754
512 links	0.031	3.081
relax iters (256 links):		
50 iters	0.522	0.569
150 iters	0.684	0.782
300 iters	0.766	1.105
600 iters	0.825	1.779
regen attempts (256 links):		
10 regen	0.684	0.782
20 regen	0.684	1.162
40 regen	0.684	1.910
XPBD step scale sr=pr (256 links, 150 it):		
x1.0 steps	0.684	0.782
x1.5 steps	0.341	0.785
x2.0 steps	0.022	0.787
x2.5 steps	0.000	0.795
margin 0.15->0.30	0.453	0.796
step x2.0 + margin0.30	0.000	0.789
step x2.0, iters=50	0.023	0.565

constant_spacing (the convergence fix):

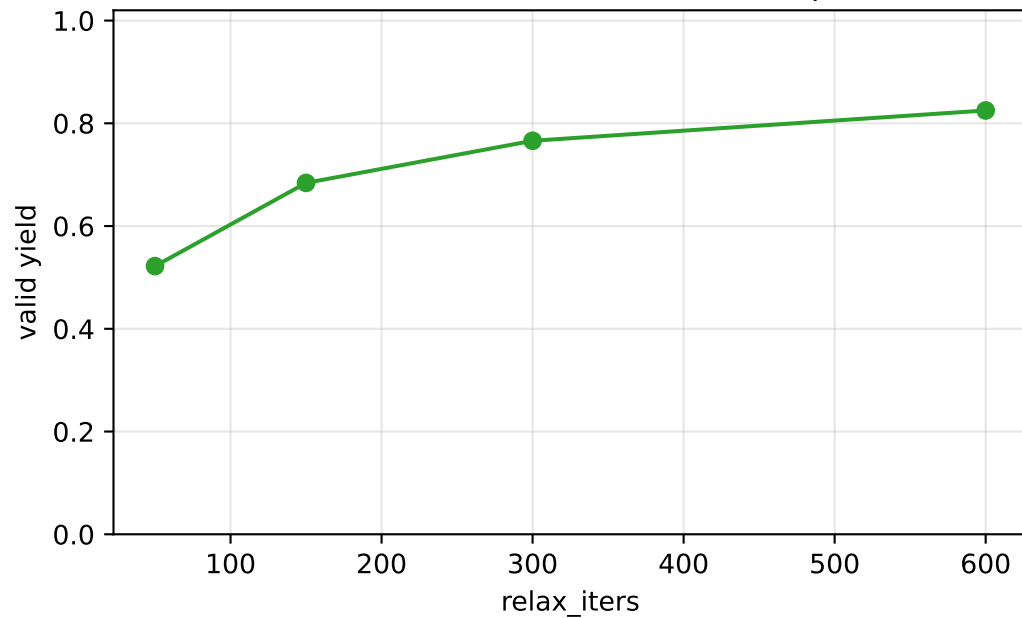
spacing=0.30 m, N_max=384 0.999 0.557 <- yield 0.684 -> 0.999

Lever impact on yield and speed (measured E=8192)

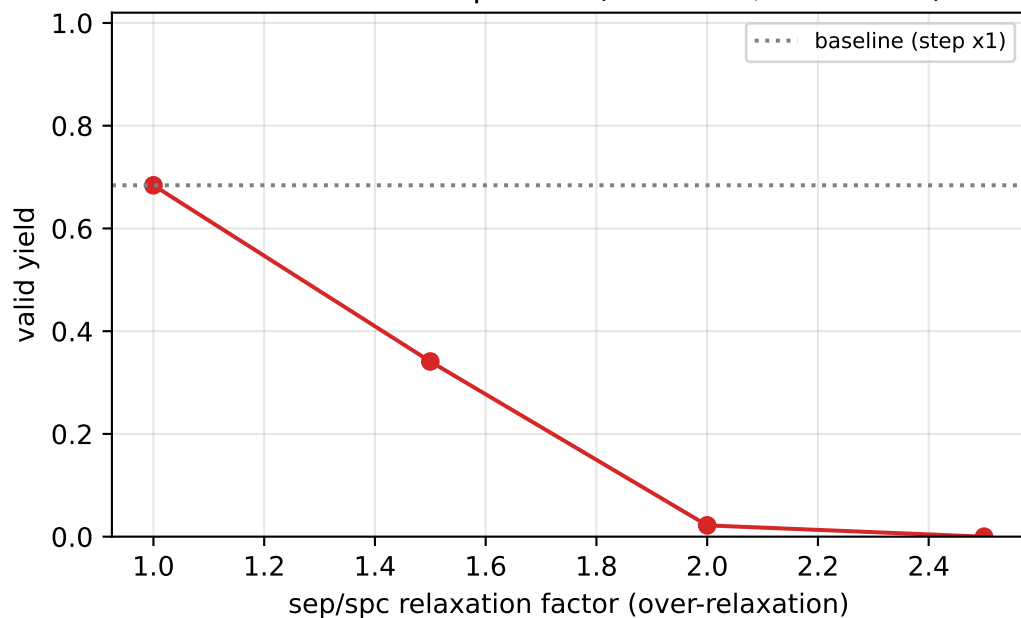
Yield vs chain links (iters=150)
(resolution-relative -- see caveats)



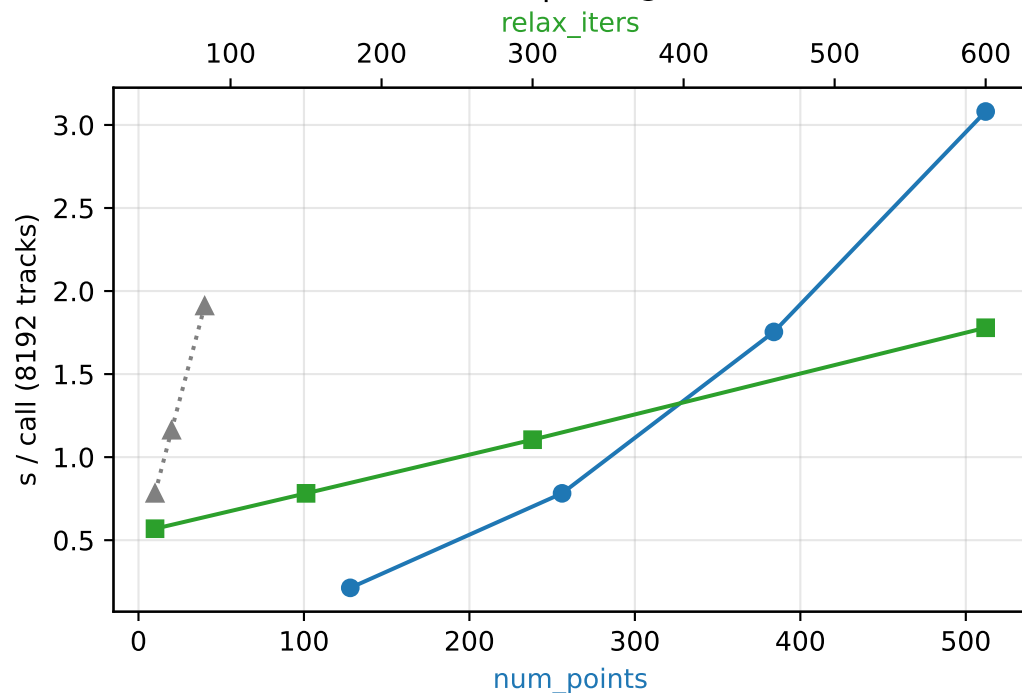
Yield vs relax iters (256 links, fair comparison)



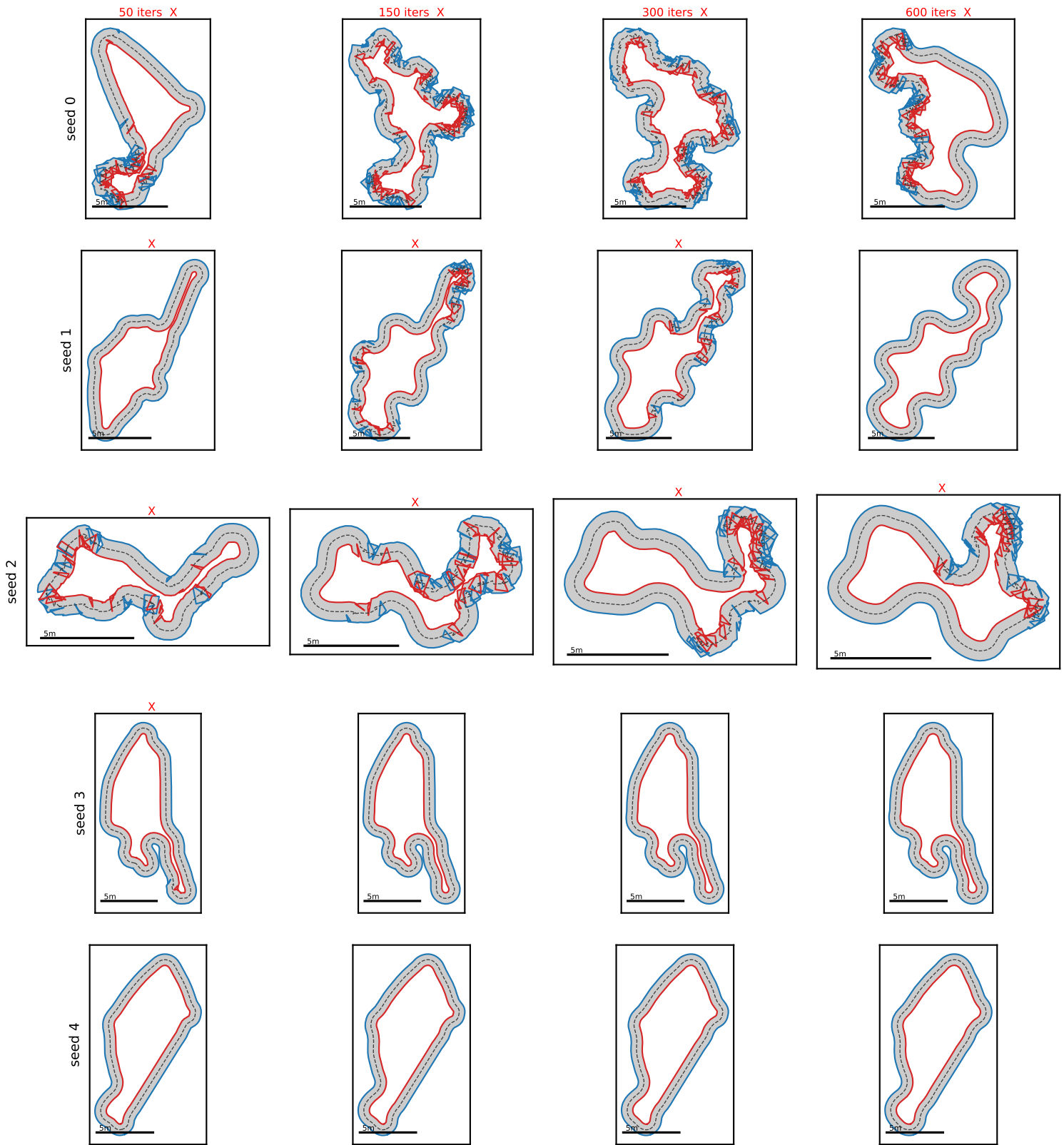
Yield vs XPBD step scale (256 links, iters=150)



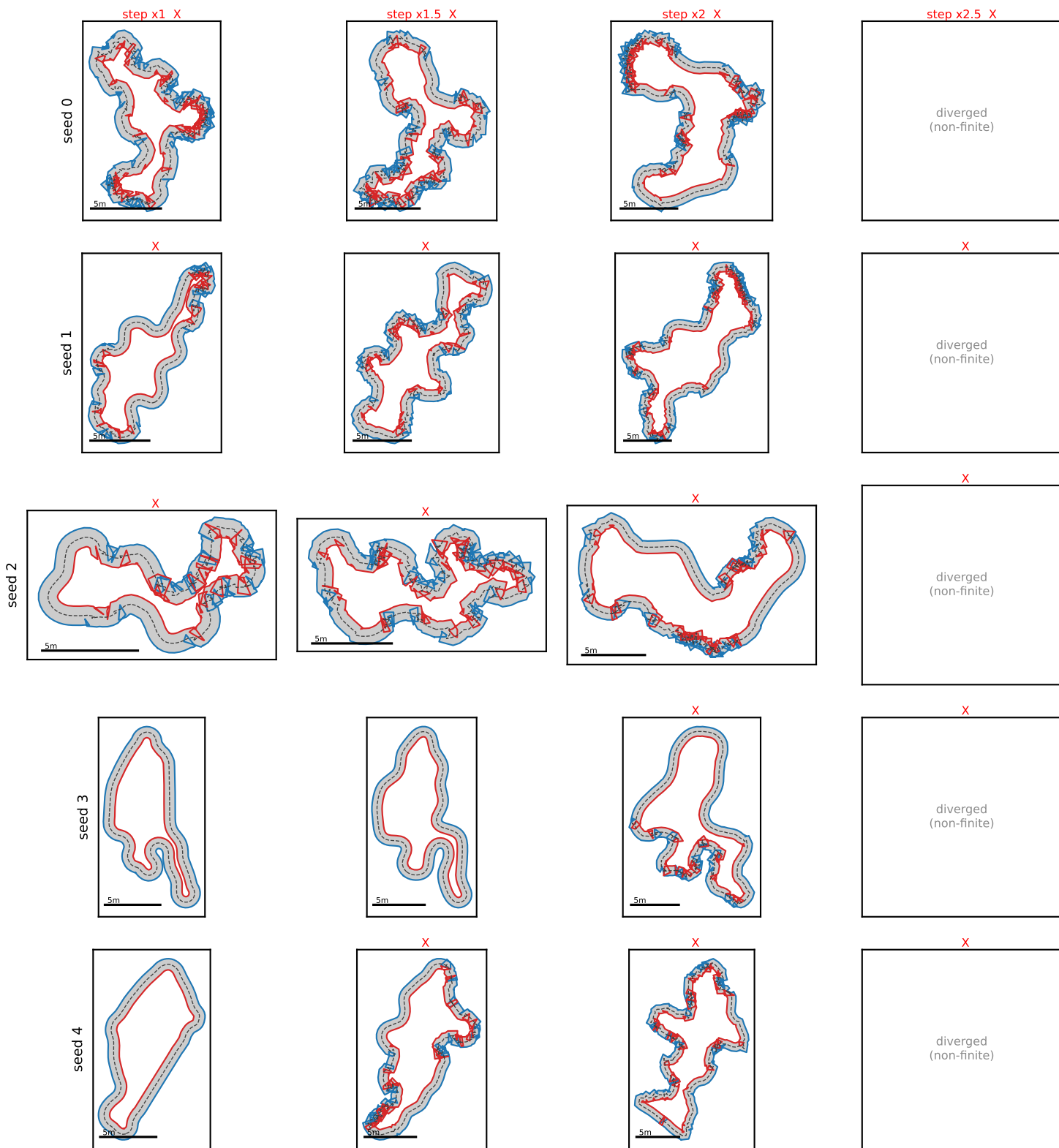
End-to-end speed @ E=8192



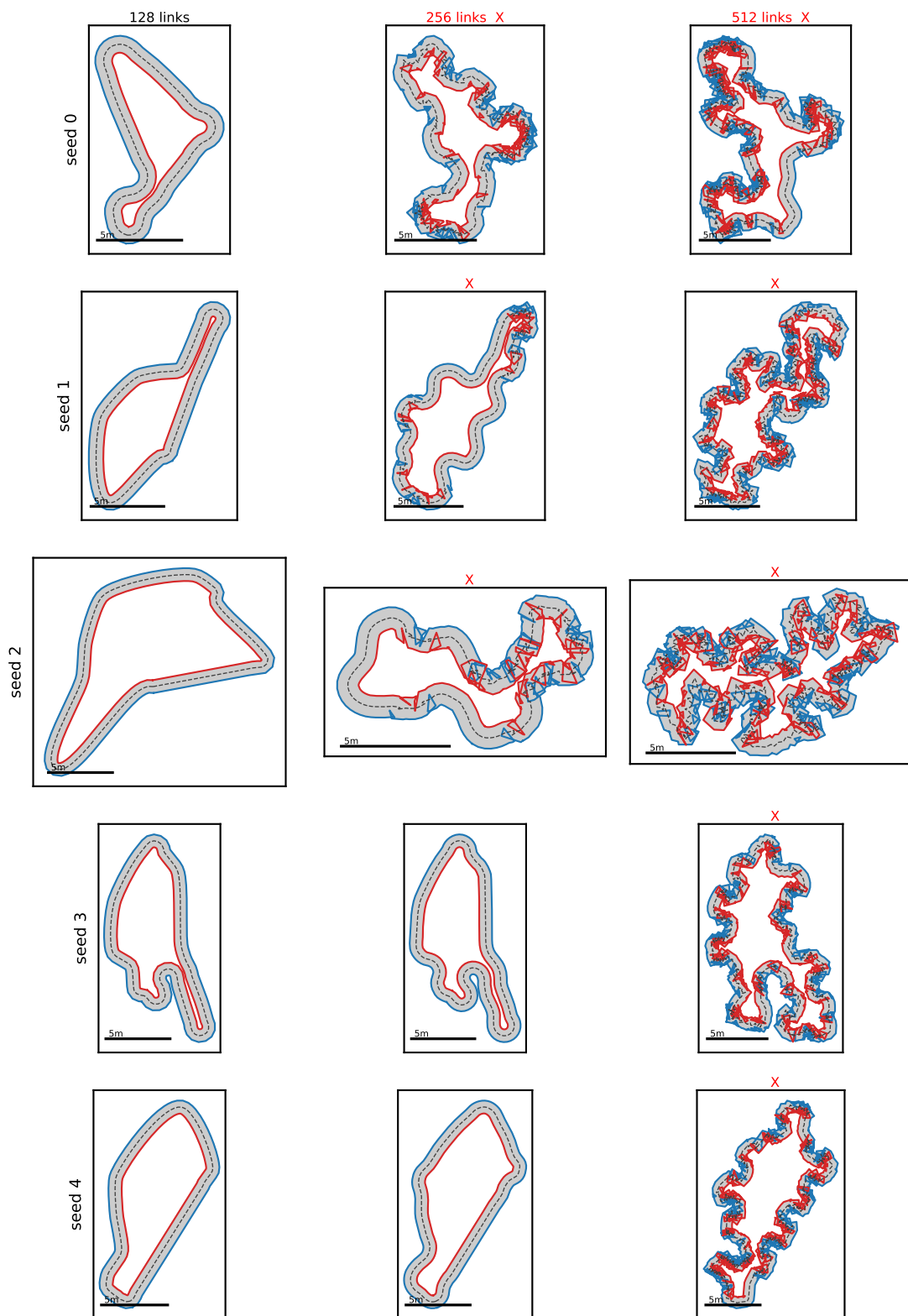
Fixed-seed impact of RELAX ITTERS (256 links)
Same generated track per row; columns relax it more. Watch pinched tracks
converge (X = invalid). Yield is convergence-limited.



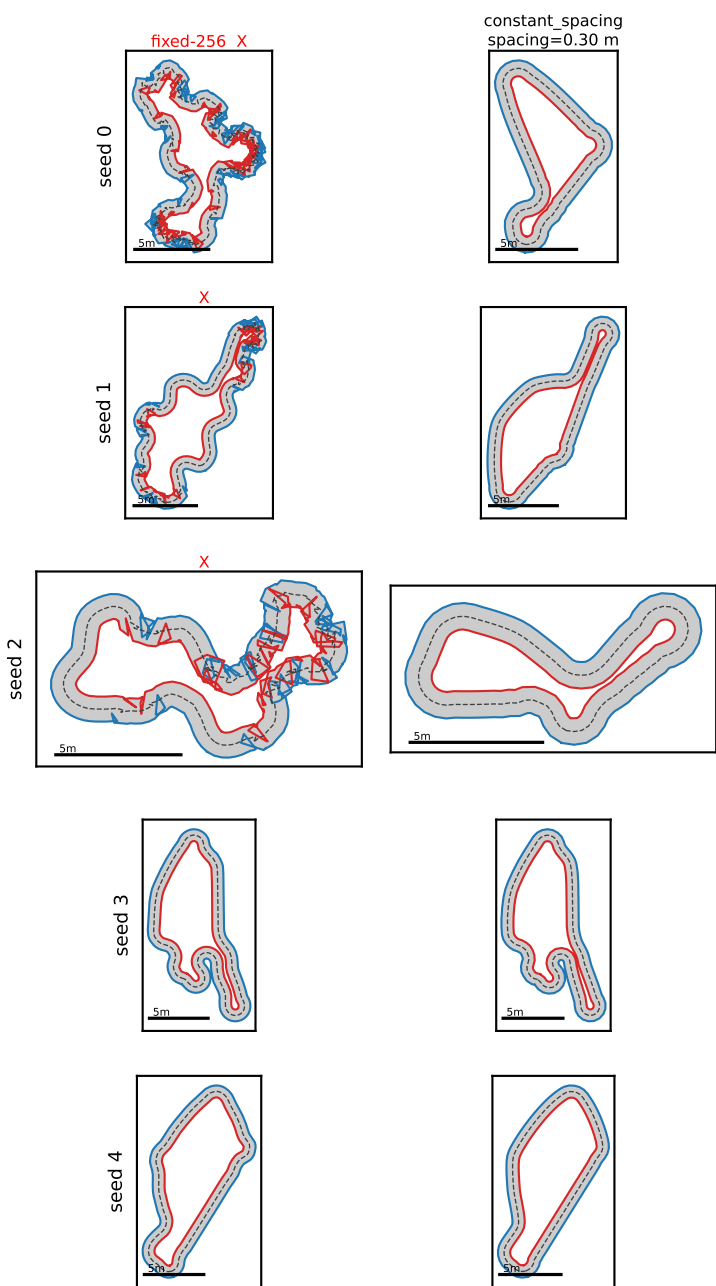
Fixed-seed impact of XPBD STEP SIZE (256 links, iters=150)
 Same generated track per row; columns use larger over-relaxation steps (sep/spc
 factor). The Jacobi solve is unstable for factor > 1: bigger steps
 OSCILLATE/diverge -> tracks degrade into invalid (X). Larger != faster here.



Fixed-seed impact of CHAIN LINKS (iters=150)
 Same seeds; more links = finer relax/validation resolution
 (note: the accepted track can differ since the gate
 resolution changes).



Constant spacing — the convergence fix
 Fixed-256 yield=0.684 (E=8192) vs constant_spacing
 spacing=0.30 m yield=0.999 (E=8192, 0.557 s/call, 35
 B). Constant spacing is the convergence fix: adaptive
 arc-length resampling to $\sim 0.6 \times \text{half_width}$ ensures the
 PBD relaxation sees a uniform, well-conditioned chain
 — eliminating the pinching that fixed-N misses at tight
 bends. Both at the default max_regen_iters=10;
 relaxation becomes lossless (valid $\approx$ generation-valid
 so generation/regen is the new ceiling, and it is
 faster at equal regen (~ 0.55 vs ~ 0.79 s/8192). X =
 invalid track.



Findings & recommendation

FINDINGS (1 m / 20 m regime, E=8192):

- * SPEED: baseline 256 links/150 it = 0.79 s/8192 (235 MB). Scales $\sim O(\text{links}^2)$ and $\sim$ linear in iters; memory is tiny (<0.5 GB of 16).
- * RELAX ITERS $\rightarrow$ genuinely raise yield (fair, fixed-256-res): 0.52/0.68/0.77/0.83 at 50/150/300/600 it. Yield is RELAXATION-CONVERGENCE-limited.
- * XPBD STEP SIZE (over-relaxation) $\rightarrow$ larger steps HURT, decisively: yield 0.68/0.34/0.02/0.00 at sep/spc factor x1/1.5/2/2.5 (speed unchanged). The solve is JACOBI (constraints applied simultaneously), so factor > 1 oscillates/diverges; x1 is already the stability ceiling. Aiming further (margin 0.15 $\rightarrow$ 0.30) also drops yield to 0.45. Faster convergence would need a better solver (Gauss-Seidel / Chebyshev / chunking), not bigger steps.
- * CHAIN LINKS $\rightarrow$ more links LOWERS native yield + costs $O(N^2)$. BUT yield is RESOLUTION-RELATIVE (the thickness curvature term is sampled at num_points), so link counts are NOT directly comparable -- fewer links = looser gate, not better tracks. Don't tune links to chase yield.
- * REGEN ATTEMPTS $\rightarrow$ NO yield change (10/20/40 all 0.684), only slower. The regen loop gates on GENERATION simplicity (saturated $\sim 100\%$); final validity is the POST-RELAX check applied after the loop, which regen can't fix.

CAVEATS (adversarially verified):

- * Yield is 'valid at the working resolution', not resolution-absolute.
- * Re-validating at finer resolution collapses the thickness CURVATURE term (artifact of resampling a polyline); SEPARATION stays honest (~ 0.47 m vs the 0.49 m target) -- the 1 m-band tracks genuinely do NOT self-overlap.

RECOMMENDATION (1 m/20 m): keep num_points=256; raise relax_iters to ~ 300 -600 for higher honest yield (0.77-0.83). Do NOT enlarge XPBD steps or margin (they destabilize), and leave max_regen_iters=10 (more is wasted). The ceiling is relaxation convergence -- the real win is a faster-converging solver.